

gene-mean $\bar{\delta}_g$ ($\alpha = 0$)



variant-specific δ_v ($\alpha = 1$)



$$\hat{\delta}(v, \alpha) = (1 - \alpha) \bar{\delta}_g + \alpha \delta_v$$

synthetic predictor family



α

0

0.25

0.5

0.75

1

P1

P2

P3

P4

P5

true quality



Interpolating between the gene-mean profile and each variant's measured effect builds predictors whose true quality order is fixed by construction.